

# Abhay Mathur

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## EDUCATION

- **DJ Sanghvi College of Engineering, Mumbai** **December 2021 - May 2025**  
B.Tech in Artificial Intelligence and Machine Learning, CGPA: 8.4

## SKILLS

- **Tech Skills:** Agentic AI, Google ADK, LangGraph, Langchain, RAG, ML, Deep Learning, GNNs, Python, Tensorflow, FastAPI
- **Non-tech Skills:** Public Speaking, Technical Writing, Singing, Playing Guitar, Piano, Table Tennis, Speedcubing

## EXPERIENCE

- **Searce - AI Engineer** **June 2025 - Present**
  - ◇ **Smart Clip Search & Video Stitching Platform**
    - Built a smart search & video compilation platform for a leading action-camera manufacturer, created and unified two pipelines (event-specific queries and thematic highlight reels) behind one interface that auto-classifies and routes each query.
    - Implemented semantic search over 127K+ video segments using Vertex AI Vector Search, filtering against user metadata and then stitching them together using ffmpeg along with Lyria-generated background music.
  - ◇ **Anomaly Diagnosis Framework**
    - Designed a multi-agent LLM system using Google ADK for automated root-cause analysis and diagnosis of network and database anomalies.
    - Developed 11 sub-agents that performed different diagnostic steps and also established correlation between the network and database anomalies.
    - Built a chatbot to interact with the anomaly database (processing 1,500 anomalies per day) on Vertex AI Agent Engine and Cloud Run.
  - ◇ **Agentic AI Data Aggregation System**
    - Built an agentic framework using Google ADK that leveraged data in Github, Benchling and GCS buckets to answer complex user queries.
    - Built a layer of custom tools on top of the Github MCP Toolset in order to paginate files above 2GB
  - ◇ **Graph RAG & Graph Analytics**
    - Developed Neo4j-based Graph RAG system by extracting entities & relationships from unstructured documents.
    - Modeled membership and case-flow data, and explored GCNs for graph representation learning.
- **Navsahyog (NGO) - AI Intern** **June 2024 - June 2025**
  - ◇ Mentored by a Principal Research SDE at Microsoft Research on an open-source project, "Jugalbandi Studio", to build and integrate a multilingual RAG-based chatbot with Telegram for the NGO.
  - ◇ Built a document translation website and a grant-writing assistant using prior proposals as contextual grounding.
- **Airrchip - AI Developer Intern** **March 2024 - August 2024**
  - ◇ Built a RAG-based chatbot for stock markets using live news, knowledge graphs, and NER analysis on stock trends.
  - ◇ Developed and deployed the backend of a medical application using FastAPI with AWS DynamoDB and S3.

## RESEARCH PAPERS

- **KAJU - Knowledge Assistant for Joint Utility** **IEEE Link**
  - ◇ Architected and developed a multi-agent LLM system (FastAPI + Java GUI) enabling autonomous code modification, debugging, and system command execution.
  - ◇ Implemented RAG-based project integration using ChromaDB for context-aware code retrieval and intelligent CRUD operations on large codebases.
  - ◇ Achieved 88.75% Task Success Rate and 93.45% recovery rate across 124 automated installation evaluations.

## OPEN SOURCE CONTRIBUTIONS

- **Microsoft - AG2 (AutoGen)** **February 2024**
  - ◇ Fixed an issue in AutoGen's User Proxy Agent that prevented automatic shell script execution by updating PowerShell command handling to support newer PowerShell versions, enabling compatibility across operating systems.

## ACHIEVEMENTS

- Searce Inc. Agentic AI Hackathon 2025 First Place (Project Link)
- IIT Bombay Techfest 2023 Hack-AI Hackathon Second Runner Up (Project Link)
- IIT Roorkee Sociothon 2023 Hackathon Second Runner-up (Project Link)
- Datahack 2.0 Hackathon Winner in Problem Statement (Project Link)